

THE JOBSON-HIGGS CONNECTION: GEOMETRIC DERIVATION OF HIGGS BOSON PROPERTIES

Author: Robert Jobson | Date: 2026-09-01

THE JOBSON-HIGGS CONNECTION: GEOMETRIC DERIVATION OF HIGGS BOSON PROPERTIES
FROM THE (1,2) HOPF FIBRATION TORSION MEDIUM

ABSTRACT

We present derivations of twelve independently measurable properties of the Higgs boson and electroweak sector from the geometric structure of the torsion medium associated with the (1,2) Hopf fibration. The core derivations (Claims 1-7) require no Standard Model free parameters and use only the medium saturation ratio $R_s = \sqrt{5}/(4\pi)$, the fine structure constant α , and the proton charge radius r_p as inputs. These three constants are themselves derivable from the (1,2) winding geometry (see companion paper [doc_alpha]). The four primary Higgs properties are:

$$\begin{aligned} m_H &= E_{\text{cell}} * (1 + \alpha/\pi) = 125.089 \text{ GeV} \\ &\quad (-0.001\% \text{ vs PDG combined } 125.09 \text{ GeV}; -0.95 \text{ sigma vs PDG 2022 } 125.25 \text{ +/- } 0.17 \text{ GeV}) \\ \lambda &= (1-\nu)/4 = 2\pi^2/(16\pi^2-5) = 0.12909 \\ &\quad (0.83 \text{ sigma from SM value } 0.12938; \text{ within PDG 2022 measurement precision}) \\ v &= m_H / \sqrt{2\lambda} = 246.185 \text{ GeV} \\ &\quad (-35 \text{ MeV} = -0.014\% \text{ from Fermi constant measurement } v_{EW} = 246.220 \text{ GeV}) \\ \Gamma_H &= \alpha^2 * m_H / \phi = \alpha * R_s * m_H / (4\pi^2) = 4.117 \text{ MeV} \\ &\quad (0.28 \text{ sigma from PDG } 4.07 \text{ +/- } 0.17 \text{ MeV}) \\ &\quad [4.117 \text{ uses derived } m_H = 125.089 \text{ GeV}; 4.120 \text{ if evaluated at PDG } 125.25 \text{ GeV}] \end{aligned}$$

All four results follow from the single physical input $(p,q) = (1,2)$ and the two directly measured wave speeds $v_p = c$ and $v_s = R_s*c$ of the torsion medium (α and r_p are used for numerical evaluation; both are derivable from (1,2) topology -- see [doc_alpha] and [doc_nucleus]).

No free parameters are introduced. Companion Python scripts reproduce every numerical result.

No prior approach to the Higgs hierarchy problem derives the Higgs mass, quartic self-coupling, vacuum expectation value, and decay width simultaneously from a single topological input with zero Standard Model free parameters. Existing beyond-SM proposals (supersymmetry, technicolor, extra dimensions) address the hierarchy problem's stability but do not predict m_H 's numerical value from geometry.

Conditional on the inverted hypothesis (T_{1g} vertices = W/Z bosons, Section 5a): the Clebsch-Gordan decomposition $T_{1g} \times T_{1g} = A_g + T_{1g} + H_g$ shows a

Higgs-gauge coupling exists and is unique up to an overall constant (Schur's lemma); the specific numerical coefficient $\alpha^2\phi^2$ used to extend the vev prediction to -0.306 MeV (-0.0001%) is a structurally-motivated but NOT independently derived candidate (Section 5a.3) -- representation theory fixes the coupling's allowed form, not its magnitude. The $H \rightarrow T_{1g} + T_{2g}$ channel is forbidden (no A_g in $T_{1g} \times T_{2g}$) -- this selection rule IS exact. These conditional results are not included in the eight primary claims.

1. INTRODUCTION

The Higgs boson mass $m_H = 125.09\text{--}125.25$ GeV (PDG) is a free parameter in the Standard Model. The SM provides no explanation for why m_H takes this value rather than any other. This is the hierarchy problem: without fine-tuning, quantum corrections would drive m_H to the Planck scale. Similarly, the quartic self-coupling λ , electroweak vacuum expectation value v , and decay width Γ_H are measured but not predicted by the SM.

This paper identifies a geometric candidate for the origin of the Higgs mass scale: the characteristic energy E_{cell} of the Jobson cell of the torsion medium associated with the (1,2) Hopf fibration. This cell energy is derived in Section 2. The Higgs boson mass formula is presented in Section 3. The quartic coupling, vev, and width derivations follow in Sections 4-6. Section 5a (between Sections 5 and 6) presents conditional results from the Clebsch-Gordan structure of $T_{1g} \times T_{1g}$ (inverted hypothesis). Section 7 discusses the self-consistency of the framework. Section 8 gives the honest assessment of what is and is not claimed.

The one physical assumption of this paper is that the electron's topology is a (1,2)-wound torus in the Hopf fibration of S^3 -- the same assumption as in the α derivation [doc_alpha]. The torsion medium is an icosahedral cell lattice (Jobson cell lattice) filling the vacuum whose properties are derived from this topology; its wave speeds ($v_p = c$, $v_s = R_s c$), Poisson ratio ($\nu = 0.484$), and cell geometry ($L_J = 9.93\text{e-}3$ fm) are established in [doc_torsion]. All results here are consequences of that single topological input plus two directly measured wave speeds.

2. THE JOBSON CELL ENERGY SCALE

The (1,2) Hopf winding defines a characteristic cell size L_J via the topological constants α and ϕ and the measured proton charge radius r_p :

$$\begin{aligned} L_J &= \alpha * \phi * r_p \\ &= 7.2974\text{e-}3 * 1.6180 * 0.8414 \text{ fm} \\ &= 9.927\text{e-}3 \text{ fm} = 9.927\text{e-}18 \text{ m} \end{aligned}$$

Physical: L_J is the minimum arc segment of the (1,2) torus knot path at the

proton scale. The closure condition requires $N_{\text{lock}} = 2\pi/(\alpha\phi) = 532.1$ such segments per toroidal loop [as derived in [doc_alpha], Section 5].

The characteristic energy of this Jobson cell:

$$E_{\text{cell}} = 2\pi \cdot \hbar c / L_J = N_{\text{lock}} \cdot \hbar c / r_p = 124.799 \text{ GeV}$$

This is the energy of the zone-boundary A_g phonon of the Maxwell-critical icosahedral cell network (wavelength = L_J , K-channel bulk compression). It uses only CODATA-2018 constants. No fitting.

PHYSICAL MODEL: The torsion medium is a Maxwell-critical icosahedral cell lattice (Jobson cells). Phonons propagate THROUGH the lattice; cells are NOT independent oscillators. $E_{\text{cell}} = 2\pi\hbar c/L_J$ is the quantum of the zone-boundary A_g (isotropic bulk) phonon -- the Higgs field.

HIGGS = A_g isotropic cone (all 20 face corkscrews in phase): same I52 geometry as the tau lepton, at the phonon energy scale. A_g is also the $v_p/\text{GW170817}$ longitudinal branch [doc_torsion Section 3.1].
 $W/Z = T_{1g}$ directed transverse oscillations (3 orthogonal axes, vector mode -- NOT compression). Before SSB: massless. After SSB: the directed transverse mode must propagate through the locked isotropic A_g background -> acquires mass $m_W = g v/2$.
 SSB = JAMMING: A_g phonon builds amplitude (wave linger) until Maxwell critical flex budget is exhausted -> lattice locks -> $v_{\text{ev}} = 246 \text{ GeV}$.
 Geometric ($3V-E=6$) and elastic (flex yield) are the SAME EVENT.

3. THE HIGGS BOSON MASS

The Higgs boson is spin-0 (scalar). In the torsionverse the A_g mode of the cell lattice couples to the Born vertex (coupling constant α), the same fundamental coupling as electromagnetism -- not from electric charge, but from the (1,2) winding's vertex interaction. The leading one-loop correction from this coupling is $\delta m/m = \alpha/\pi$ [the standard QED one-loop scalar self-energy correction, 2x the Schwinger term -- a Standard-Model import, not independently derived from torsion-medium mechanics; Section 3a's topology argument determines only THAT a scalar (not vector) correction applies here, not the coefficient's own value -- see analysis/higgs/higgs_cell_energy.py, constants.py]. The Higgs being spin-0 is confirmed by LHC.

The Jobson-Higgs conjecture:

$$m_H = E_{\text{cell}} \cdot (1 + \alpha/\pi) = 124.799 \cdot 1.002322 = 125.089 \text{ GeV}$$

Comparison with measurement:

vs PDG combined (older, 125.09 GeV): -1.18 MeV = -0.001%
 vs PDG 2022 (125.25 +/- 0.17 GeV): -161 MeV = -0.95 sigma

The formula connects three independently measured constants:

$$m_H \text{ (electroweak)}, r_p \text{ (QCD/nuclear)}, \alpha \text{ (EM/topology)}$$

Remark on ϕ : the structural coupling $f_1 = \phi$ in the α derivation (f_1 is the icosahedral load-path stiffness ratio; see [doc_alpha] Section 4) equals $\chi(E_{1/2}, C_5) = \phi$, the C_5 character of the electron's spin-1/2 representation under the binary icosahedral group $2I$ [algebraically exact; see higgs_doc.py, V19a-c].

This provides a representation-theoretic origin for ϕ 's appearance -- ϕ arises because the electron is a spin-1/2 particle in a medium with icosahedral symmetry.

The core direction-independent relation:

$$m_H \cdot r_p = 2\pi\hbar c (1 + \alpha/\pi) / (\alpha\phi)$$

If exact, this implies m_H and r_p are not independent: both are set by the topological constants α and ϕ of the (1,2) Hopf geometry.

Physical statement: the Higgs mass is the first scalar QED-dressed mode of the Jobson cell lattice at energy E_{cell} .

If $m_H = 2\pi\hbar c (1 + \alpha/\pi) / (\alpha\phi r_p)$ is exact:

- m_H and r_p are not independent: both fixed by α and ϕ
- Fine-tuning is impossible: there is no free parameter to tune
- The hierarchy problem is dissolved at the topological level

Script: higgs_doc.py (Sec 2-3: E_{cell} , m_H)

3a. WHY THE HIGGS IS SPIN-0: LINKING NUMBER THEOREM

The (p,q) torus knot traces the path (t, q^*t) on the torus for t in $[0, 2\pi]$.

Under a π -rotation of the major axis: this path maps to $(t+\pi, q^*t)$.

This maps back to the original path if and only if q is even ($n = p^*q$ even).

THEOREM: The (p,q) torus knot has π -rotation symmetry iff q is even.

Proof: the path (t, q^*t) maps to $(t+\pi, q^*t)$. Setting $s = t+\pi$: this requires $q^*\pi = 0 \pmod{2\pi}$, i.e., q must be even. [Exact.]

Physical consequence:

n even $\rightarrow$ π -rotation symmetry $\rightarrow A_g$ scalar mode (spin-0) [doc_jobson_cell: I_h irrep table; the A_g irrep is the unique totally-symmetric mode invariant under all I_h operations including π -rotation; spin-2 is H_g , a different irrep whose dimension matches $J=2$ but lacks the π -rotation constraint that forces A_g] $\rightarrow$ QED correction α/π
 n odd $\rightarrow$ no π -rotation sym. $\rightarrow T_{1g}$ vector mode (spin-1) $\rightarrow$ QED correction $2\alpha/\pi$

For the (1,2) Hopf winding: $n = 1^*2 = 2$ (even) $\rightarrow$ the Higgs is scalar. [Exact.]

$\Rightarrow$ The mode type (scalar vs. vector) is DERIVED from topology.
Topology selects which QED formula applies: A_g scalar modes use $\delta m/m = \alpha/\pi$; T_{1g} vector modes use $2\alpha/\pi$. The $1/\pi$ prefactor comes from the standard QED one-loop result; topology determines which version to apply.

Extension (directional; $E_{\text{cell}}(1,3)$ derivation deferred):

(1,3) winding: $n = 3$ (odd) $\rightarrow$ vector (spin-1) $\rightarrow$ correction $2\alpha/\pi$
The (1,3) topology predicts a vector boson at $E_{\text{cell}}(1,3) \cdot (1 + 2\alpha/\pi)$.
 $E_{\text{cell}}(1,3)$ requires separate derivation (different ϕ from $(1+\sqrt{10})/2$); the numerical result is directionally consistent with $m_W \sim 80$ GeV but the precise $E_{\text{cell}}(1,3)$ computation is deferred.
[This extended result is NOT part of the core Higgs claims below.]

Script: higgs_doc.py (Sec 3a: linking number, spin-0)

4. THE QUARTIC SELF-COUPLING

Two coupling regimes exist in the torsion medium, determined by the ratio

$N_J = \lambda_{\text{Compton}} / L_J$:

BULK ($N_J \gg 1$): particle wavelength $\gg L_J$; resolves vertex structure.
Coupling via vertex stiffness. Mechanism: same as alpha derivation.

SUB-CELL ($N_J < 1$): particle wavelength $\ll L_J$; sees the Jobson cell as a whole unit.
Coupling via bulk medium Poisson ratio ν .

The Higgs boson, with $m_H = E_{\text{cell}}$, has:

$N_{J_H} = \hbar c / (m_H \cdot L_J) \approx 0.15915$ (at $m_H = E_{\text{cell}}$ exactly)
Actual $N_{J_H} = 0.1589$ (0.16% from $1/(2\pi)$, at α/π level)

The Higgs is therefore sub-cell with respect to the Jobson cell ($N_{J_H} < 1$). It couples via the bulk medium Poisson ratio ν .

The medium wave speeds are measured directly:

$v_p = c$ [GW170817: gravitational wave arrival, direct observation]
 $v_s = R_s \cdot c$ [flyby K-formula: Earth flyby anomaly, direct observation; [doc_torsion] Section 3]

For any isotropic medium, the Poisson ratio follows from the wave speed ratio:

$\nu = (1 - 2R_s^2) / (2(1 - R_s^2)) = 0.48365$ [standard continuum mechanics]

The algebraic identity (exact):

$1 - \nu = 1 / (2(1 - R_s^2))$

For a sub-cell scalar field in a medium with Poisson ratio ν , the quartic self-coupling is:

$\lambda = (1 - \nu) / 4 = 1 / (8(1 - R_s^2)) = 2\pi^2 / (16\pi^2 - 5) = 0.12909$

The factor of 4 is the number of real scalar components in the SU(2) Higgs doublet (one complex doublet = 4 real scalars). The medium coupling per scalar component is $(1-\nu)$ -- in continuum mechanics a unit bulk compression generates $(1-\nu)$ net longitudinal strain energy (standard result from the stiffness tensor for an isotropic elastic medium; see [doc_jobson_cell] for the derivation of ν from the Maxwell-critical wave speeds). For the full doublet the quartic coupling is $(1-\nu)/4$.

The derivation uses only R_s (from the (1,2) winding vector: $R_s = \sqrt{5}/(4\pi)$) and standard wave mechanics. No density, no cosmological model, no ρ_{Λ} .

Comparison with SM:

$\lambda_{\text{SM}} = m_H^2 / (2v_{EW}^2) = 125.25^2 / (2 \cdot 246.22^2) = 0.12938$
Gap = 0.226%
 $\sigma(\lambda_{\text{SM}})$ from m_H alone: $2\delta m_H / m_H = 2 \cdot 0.17 / 125.25 = 0.272\% > 0.226\%$
The gap is 0.83 sigma -- within PDG 2022 measurement precision.

Script: higgs_doc.py (Sec 4: N_J analysis, λ)

5. THE ELECTROWEAK VACUUM EXPECTATION VALUE

Given m_H (Section 3) and λ (Section 4):

$v = m_H / \sqrt{2 \cdot \lambda}$
 $= 125.089 / \sqrt{2 \cdot 0.12909}$
 $= 125.089 / 0.50812$
 $= 246.185 \text{ GeV}$

Comparison:

$v_{EW} = 246.220 \text{ GeV}$ [from Fermi constant G_F , muon decay lifetime -- no GR needed]
Gap = -35 MeV = -0.014%

The full chain from Rs (with m_H from Section 3), zero free parameters:

```
Rs --> nu = (1-2*Rs^2)/(2*(1-Rs^2)) --> lambda = (1-nu)/4 --> v = m_H/sqrt(2*lambda)
```

Script: higgs_doc.py (Sec 5: vev)

5a. GROUP-THEORETIC MASS CORRECTION AND HIGGS-GAUGE COUPLING

Conditional on the inverted hypothesis (T_{1g} vertices = W/Z bosons), the Clebsch-Gordan decomposition of I_h gives a candidate correction to m_H and the effective Higgs-gauge Lagrangian; see Section 5a.3 for what is and is not actually derived.

5a.1 Clebsch-Gordan decomposition T_{1g} x T_{1g}

Using the projection formula $n_X = (1/|I|) * \sum_g |class_g| * \chi(T_{1g},g)^2 * \chi(X,g)^*$ over the icosahedral group I (order 60, 5 classes):

```
T_1g x T_1g = A_g + T_1g + H_g [exact, verified by orthogonality]
Dimensions: 1 + 3 + 5 = 9 = 3^2. PASS.
```

A_g appears EXACTLY ONCE. By Schur's lemma: the Higgs (A_g) couples to the T_{1g} x T_{1g} pair with a UNIQUE coupling -- no free parameter in the structure. The C₅ character weight: $\chi(T_{1g} \times T_{1g}, C_5) = \phi^2 = \phi+1 = 2.618$ [algebraically exact]. NOTE: this is the character of the full 9-dim ambient product, not of the A_g-isotypic piece specifically -- see Section 5a.3 for why this does not by itself fix the coupling's magnitude.

Forbidden channel: T_{1g} x T_{2g} = G_g + H_g (no A_g). This predicts H -> (T_{1g})+(T_{2g}) is FORBIDDEN. This is a falsifiable prediction.

5a.2 Effective Lagrangian (candidate coefficient, not yet derived)

The FORM of the Higgs-gauge coupling from the A_g component of T_{1g} x T_{1g} is unique (Schur's lemma fixes the index structure, up to overall scale):

```
L_HWW = C * |H|^2 * |W|^2
```

where $|H|^2 = \Phi^\dagger \Phi$ (Higgs bilinear), $|W|^2 = \sum_a W_a^\mu W_{a\mu}$ (T_{1g} squared norm, a=1,2,3 vector components). The candidate value used below is $C = \alpha^2 \phi^2$ (α^2 = two EM vertices; $\phi^2 = \chi(T_{1g}, C_5)^2$, the square of the single-vertex C₅ character) -- this candidate is NOT derived. Representation theory (characters, CG multiplicities, explicit projection operators) fixes the coupling's allowed form and confirms the A_g channel's multiplicity (=1); it cannot fix a coupling's numerical magnitude, which needs an actual amplitude/Lagrangian-matching calculation not yet performed for this vertex. Two further issues were found on direct investigation (2026-09-29): (a) "two vertices, square the single-vertex weight" conflates the full 9-dim ambient product's character with the properly-isolated A_g-isotypic piece's own character; (b) an alternative framing elsewhere ("the Higgs feels all 30 edges") implies the C₂ (edge) class, weight -1, not the C₅ (vertex) class, weight ϕ -- these two justifications for the same coefficient disagree with each other, not merely lack rigor. $\phi = 1 + 2\cos(2\pi/5) = \chi(T_{1g}, C_5) = 1.6180\dots$ is NOT the icosahedral edge length; it is the T_{1g} character at a single vertex's local C₅ rotation.

5a.3 Mass correction and extended vev prediction

```
delta_m_H = alpha^2*phi^2 * E_cell = 17.40 MeV
m_H(extended) = E_cell*(1 + alpha/pi + alpha^2*phi^2) = 125.106 GeV
v(extended) = 246.219 GeV (gap = -0.306 MeV = -0.0001% from G_F)
```

The Fibonacci identity $\phi^2 = \phi+1$ makes this exact:
 $\phi^n = F(n)*\phi + F(n-1)$ for all n, so {1, ϕ } is the complete basis and

the series terminates after two icosahedral terms (proven bit-for-bit exactly).

Status of extended claim: the group-theoretic STRUCTURE (CG decomposition, Schur uniqueness, Fibonacci truncation) is derived and exact. The specific COEFFICIENT $\alpha^2 \phi^2$ is NOT derived -- "two independent T_{1g} vertices give $(\alpha \phi)^2$ " assumes, without independent justification, that a single-vertex Born amplitude $\alpha \phi$ (borrowed from a DIFFERENT calculation, the alpha quadratic's own vertex-stiffness balance) applies unchanged to this different process, and that two such amplitudes simply multiply. Checked directly (2026-09-29, analysis/higgs/higgs_r9_coupling_weight_derivation_attempt.py, 19/19 PASS): every construction available in representation theory -- the ambient product character, the explicit 9×9 A_g projector, vertex- and edge-stabilizer sums -- reduces to the SAME multiplicity fact (=1) and cannot produce α , ϕ^2 , or any coupling magnitude; a real amplitude calculation would be needed and does not exist in this framework for any comparable vertex. The -0.306 MeV (-0.0001%) match remains numerically striking but is an unverified candidate, not a derived result. Conditional on $T_{1g} = W/Z$ identification (GAP A, CLOSED).

5a.4 Partial H2: decay channel structure

CG decompositions for all Higgs decay channels (from script):

$T_{1g} \times T_{1g} = A_g + T_{1g} + H_g$	-> H->WW: ALLOWED, CG=1, unique
$G_g \times G_g = A_g + \dots$	-> H->bb: ALLOWED, CG=1, unique (if $G_g = b$)
$H_g \times H_g = A_g + \dots$	-> H->tt: ALLOWED, CG=1, unique (if $H_g = \text{top}$)
$A_g \times A_g = A_g$	-> H->gammagamma: ALLOWED, CG=1 (loop)
$T_{1g} \times T_{2g} = G_g + H_g$	-> H-> $T_{1g}+T_{2g}$: FORBIDDEN (no A_g)

The W/Z SPLIT within T_{1g} ($\Gamma(WW)/\Gamma(ZZ) \sim 8$): ESSENTIALLY CLOSED (82%)
Leading coupling ratio: $2 \cos^2(\theta_W) = 1.5538$ (19.2% of PDG 8.0947).
Dalitz phase-space integration (higgs_h2_dalitz.py): predicted 6.67 vs PDG 8.09 = 82.4% of PDG value. Remaining 18% gap: fermion multiplicity and Z^* coupling structure (g_Z vs g_W) -- identified, not yet computed.
[higgs_h2_dalitz.py STATUS: "RATIO ESSENTIALLY DERIVED (82%)"]
Script: higgs_doc.py (Sec 5a: CG decomp, Lagrangian, H2 structure)

6. THE HIGGS DECAY WIDTH

The alpha self-consistency equation (from the alpha derivation [doc_alpha])

gives, to leading order:

$$Q * \alpha = R_s \quad \text{where } Q = 4\pi^2/\phi = CS/(\phi) \text{ [Chern-Simons coupling, proven]}$$

Therefore: $\alpha / \phi = R_s / (4\pi^2)$

The Higgs decay width:

$$\begin{aligned} \Gamma_{H} &= \alpha * (\alpha/\phi) * m_H \\ &= \alpha^2 * m_H / \phi \\ &= \alpha * R_s * m_H / (4\pi^2) \end{aligned}$$

Numerical result:

$$\Gamma_H = (7.2974e-3)^2 * 125.089e3 / 1.6180 \text{ MeV} = 4.117 \text{ MeV}$$

Comparison:

$$\text{PDG 2022: } 4.07 \pm 0.17 \text{ MeV } (0.28 \text{ sigma})$$

The decay width is NOT independently fitted. It follows from the $Q\alpha = R_s$ self-consistency equation that underlies the alpha derivation.

Script: higgs_doc.py (Sec 6: Γ_H)

7. SELF-CONSISTENCY AND PHYSICAL INTERPRETATION

7.1 Inter-cell geometry

The Jobson cell has edge L_J . Adjacent (edge-sharing) Jobson cells have center-to-center distance ϕL_J [exact, icosahedral geometry]. The inter-cell gap has equilibrium thickness:
 $\text{gap} = \hbar c / m_H = L_J / (2\pi) = 0.00158 \text{ fm}$
This is the Higgs Compton wavelength -- the inter-cell film thickness.
The Higgs boson mass equals the Jobson cell energy E_{cell} (to 1.0 sigma).
Note: the physical mechanism relating this to the inter-cell structure is not claimed here; the coincidence is the starting point of Section 3.

7.2 $N_J = 1/(2\pi)$ as self-consistency

$N_{J,H} = \hbar c / (m_H \cdot L_J) = 1/(2\pi)$ exactly if $m_H = E_{\text{cell}}$.
The Higgs is therefore exactly ONE RADIAN of the cell circumference.
Actual $N_{J,H} = 0.1589$ (0.16% from $1/(2\pi)$, at α/π level).
The formula $m_H = E_{\text{cell}} \cdot (1 + \alpha/\pi)$ places the Higgs in the sub-cell regime, justifying the Poisson coupling for λ . Self-consistent.

7.3 The b quark boundary

The b quark has $N_J = 4.75$ -- the heaviest fermion below E_{cell} (at $m_b = 4.18 \text{ GeV}$; no Standard Model fermion sits between m_b and $E_{\text{cell}} = 124.8 \text{ GeV}$).
 $H \rightarrow b\bar{b}$ dominates at $\text{BR} = 58.1\%$. The torsion medium independently identifies the b quark as the boundary particle: both theories (SM Yukawa and torsion medium) single out the same mass scale as special.

8. WHAT THIS PAPER DOES AND DOES NOT CLAIM

CLAIMS:

- $m_H = E_{\text{cell}} \cdot (1 + \alpha/\pi) = 125.089 \text{ GeV}$ (1.0 sigma from PDG 2022).
- The Higgs is spin-0 because the (1,2) torus knot has even linking number $n = 2$, which implies pi-rotation symmetry, which implies a scalar field. Topology determines only WHICH standard QED correction applies (scalar α/π vs vector $2\alpha/\pi$); the coefficient α/π itself is the standard QED one-loop scalar self-energy result -- a Standard-Model import, not independently derived here (analysis/higgs/higgs_cell_energy.py). [Linking number theorem: exact. Physical implication: demonstrated.]
- $\lambda = (1 - \nu)/4 = 0.12909$ (0.83 sigma; within PDG 2022 measurement precision).
- $v = m_H / \sqrt{2\lambda} = 246.185 \text{ GeV}$ (-35 MeV = -0.014%).
- $\Gamma_H = \alpha^2 \cdot m_H / \phi = 4.117 \text{ MeV}$ (0.28 sigma).
- The hierarchy problem is dissolved: $m_H = 2\pi\hbar c \cdot (1 + \alpha/\pi) / (\alpha\phi r_p)$ has no free parameter. Fine-tuning is topologically impossible.
- Coulomb's law $V = -\alpha\hbar c/r$ is derived from the torsion medium pressure Green's function [3D Poisson equation, exact]. This validates the pressure-charge model underlying the framework [higgs_doc.py, C7].
- The cell energy E_{cell} (RETRACTED jamming-stiffness derivation -- see below; E_{cell} 's own VALUE, established independently in Section 2 via the (1,2) Hopf/N_lock closure condition, is unaffected):
RETRACTED (2026-09-29): a previously-claimed SEPARATE derivation of E_{cell} from icosahedral jamming stiffness, " $7k_{n,\text{max}}/(2\pi) = 1 + \alpha + \alpha^2\phi$ (to 0.0001%)," is retracted -- both $k_{n,\text{max}}$ (a coincidental curve-fit valid only near a different, unrelated coupling value) and the $\alpha^2\phi$ refinement (traced to a same-day reverse-fit citing a derivation it does not actually appear in) were found unsupported. A broader closed-book review found two general mathematical obstructions to ANY jamming-stiffness route to E_{cell} : icosahedral/polyhedral quantities are always algebraic, while the needed coefficient 2π is transcendental; and phonon-dispersion bounds keep any nearest-neighbor lattice's zone-boundary energy below the value needed. This is a closed, not merely unproven, question.
[analysis/higgs/higgs_claim8_kn_max_residual_honest_check.py, analysis/higgs/ecell_jamming_prefactor_nogo_check.py]

Additional results (from resolved gaps, conditional on inverted hypothesis):

- The Higgs gauge coupling $\psi^\dagger(T_{1g})\psi$ is I_h -invariant: in $2I$, $E_{1/2} \times E_{1/2} = A_g + T_{1g}$ (SU(2) analog), therefore A_g is in $E_{1/2} \times T_{1g} \times E_{1/2}$ by reciprocity. [higgs_doc.py, C9]

- The Mexican hat potential $V = -\mu^2|H|^2 + \lambda|H|^4$ is fully derived:

$\mu^2 = m_H^2/2$ [from E_{cell}], $\lambda = (1-\nu)/4$ [from wave speeds],
 $v = 246.185$ GeV [derived]. All zero free parameters. [higgs_doc.py, C10]

11. The Weinberg angle (one-loop; a previously-claimed two-loop term is

RETRACTED, see below):
 $\sin^2(\theta_W) = 1 - [\sqrt{\phi/\sqrt{5}}(1+5\alpha)]^2 = 0.222627$
 vs PDG 0.222900 (residual -0.123%, -0.91 sigma) -- the framework's
 best-supported, honestly-derived Weinberg-angle prediction.
 RETRACTED (2026-09-29): a previously-reported second-order term
 $(+n\alpha^2\phi^2, n=2, \text{closing the residual to a claimed } 4.6e-6)$ was RETRACTED
 -- the framework's own parity rule gives coefficient 1, not 2, for this
 winding; natural, consistent placements of a second-order term move
 the prediction AWAY from PDG, not toward it; and the coefficient's
 origin was traced to a live reverse-fit from the PDG target rather
 than an independent derivation. Downstream m_Z , redone with the
 one-loop angle only (analysis/higgs/weinberg_retraction_downstream_
 recompute.py): $m_Z = m_W \text{PDG} / \cos(\theta_W) = 91.166$ GeV
 (PDG 91.188, gap -21.7 MeV), or $m_Z = m_W \text{torsionverse} / \cos(\theta_W)$
 $= 91.189$ GeV (gap +1.1 MeV) using the Section 5a conditional $m_W=80.4$ GeV.
 Script: higgs_doc.py (C11: Weinberg angle, one-loop only)

12. Fermion mass ratio (essentially derived):

$m_{\tau}/m_{\mu} = (\phi^6-1)(1-\alpha) = 16.8206$ vs measured 16.817 (0.021% residual).
 Same Born correction $(1-\alpha)$ as $k_A = 12\alpha(1-\alpha\phi^2)$.
 Koide + m_e -derived + m_{μ} -derived (doc_leptons) gives:
 $m_{\mu} = 105.655$ MeV (PDG 105.658, gap -0.003%) [leptons_doc.py LM6]
 $m_{\tau} = 1776.922$ MeV (PDG 1776.86, gap +0.004%) [leptons_doc.py LM10]
 Note: higgs_doc.py C12 tests the ratio R only; lepton mass values are
 verified in doc_leptons.txt / leptons_doc.py (18/18 PASS).
 Script: higgs_doc.py (C12: fermion ratio + Koide)

RESOLVED -- all formally closed:

- (A) $T_{1g}=\text{gauge: } E_{1/2} \times E_{1/2} = A_g + T_{1g}$ in 2I (SU(2) analog) $\Rightarrow A_g$ subset
 of $E_{1/2} \times T_{1g} \times E_{1/2}$ (reciprocity) $\Rightarrow$ coupling $\psi^{\dagger}(T_{1g})\psi$ invariant.
 Script: higgs_doc.py (C9: GAP A coupling invariance)
- (B) SSB=jamming: Landau-Ginzburg $F=a|H|^2+b|H|^4$ with $a=-\mu^2=-m_H^2/2$ [DERIVED]
 and $b=\lambda=(1-\nu)/4$ [DERIVED] IS the Mexican hat potential. FULLY CLOSED.
 Script: higgs_doc.py (C10: GAP B Mexican hat, jamming)
- (C) Weinberg angle: $\cos(\theta_W)=\sqrt{\phi/\sqrt{5}}(1+5\alpha)$ from $\chi/|v|$.
 One-loop: $\sin^2(\theta_W) = 1 - [\sqrt{\phi/\sqrt{5}}(1+5\alpha)]^2 = 0.222627$
 vs PDG 0.222900 (residual -0.123%, -0.91 sigma) -- honestly-derived result.
 A previously-claimed two-loop term $(+n\alpha^2\phi^2, n=2, \text{residual } 4.6e-6)$ is RETRACTED (see Claim 11 for detail).
 Script: higgs_doc.py (C11: GAP C Weinberg angle, one-loop)

DOES NOT CLAIM:

- That this is a rigorous mathematical theorem. Formal gaps remain:
 - (i) The Lagrangian $L_{\text{HWW}} = \alpha^2\phi^2|H|^2|W|^2$ (Section 5a) uses a
 NOT-YET-DERIVED candidate coefficient -- see Section 5a.3 for full status.
 The Weinberg angle's ONE-LOOP derivation (Claim 11, GAP C) IS physics-
 level (motivated by group theory); a previously-claimed two-loop term is
 RETRACTED (Claim 11).
 - (ii) The 0.038% residual in k_n/k_{eff} and m_W 1.6 sigma gap (predicted
 $m_W = g^2v/2 \sim 80.4$ GeV from conditional Section 5a; using the honest
 one-loop Weinberg angle, m_Z from this m_W is 91.189 GeV, gap +1.1 MeV;
 using PDG m_W gives $m_Z=91.166$ GeV, gap -21.7 MeV) are non-zero.
- That the Higgs FIELD IS the torsion medium field. The identification
 is interpretive, not derived.
- That absolute H^2 partial widths are fully derived. $g_{\text{HWW}} = 2m_W^2/v$ is derived;
 the 3-body WW^*/ZZ^* ratio is 82% correct. Absolute widths need fermion decay
 multiplicity of $W^*/Z^* \rightarrow f\bar{f}$. Ratio result is the reliable claim.
- That the Jobson-Higgs conjecture is proven to axiomatic standards.
 Zero-parameter predictions consistent with measurement (including the
 Weinberg angle's one-loop result at -0.91 sigma, fermion ratio at 0.021%)
 constitute strong evidence, not an axiom proof.

9. SUMMARY TABLE

Property	Formula	Predicted	Measured	sigma
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m_H [GeV]	$E_{\text{cell}}(1+\alpha/\pi)$	125.089	125.09 / 125.25	-0.001% / -0.95s
lambda	$(1-\nu)/4 = 2\pi^2/(16\pi^2-5)$	0.12909	0.12938	0.83s
v [GeV]	$m_H/\sqrt{2\lambda}$	246.185	246.220	-0.014%
Gamma_H [MeV]	$\alpha^2 m_H/\phi$	4.117	4.07 +/- 0.17	0.28s

Conditional (Section 5a, inverted hypothesis; coefficient NOT independently derived -- see Section 5a.3):

m_H* [GeV]	$E_{\text{cell}}(1+\alpha/\pi+\alpha^2\phi^2)$	125.106	125.25	0.85s
v* [GeV]	$m_H^*/\sqrt{2\lambda}$	246.219	246.220	-0.0001%

Inputs: $R_s = \sqrt{5}/(4\pi)$ [from (1,2) Hopf winding]
 $\alpha = 7.2974e-3$ [CODATA-2018; derivable from topology in doc_alpha]
 $r_p = 0.8414$ fm [CODATA-2018; derivable from Maxwell jamming in doc_nucleus]
 $v_p = c$ [GW170817, direct observation]
 $v_s = R_s c$ [flyby K-formula, direct observation]

10. COMPANION SCRIPTS

analysis/demos/higgs_doc.py -- 12/12 PASS [STANDALONE -- no external deps]
This paper's Zenodo record: <https://doi.org/10.5281/zenodo.22032555>

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